

1.10.2023

(Emily)

Post Correspondence Problem

Sequence
→

$$X = \{ x_1, x_2, x_3 \} = \{ 0, 1, 011 \}$$

→

$$Y = \{ y_1, y_2, y_3 \} = \{ 1, 011, 0 \}$$

Corresponding elements

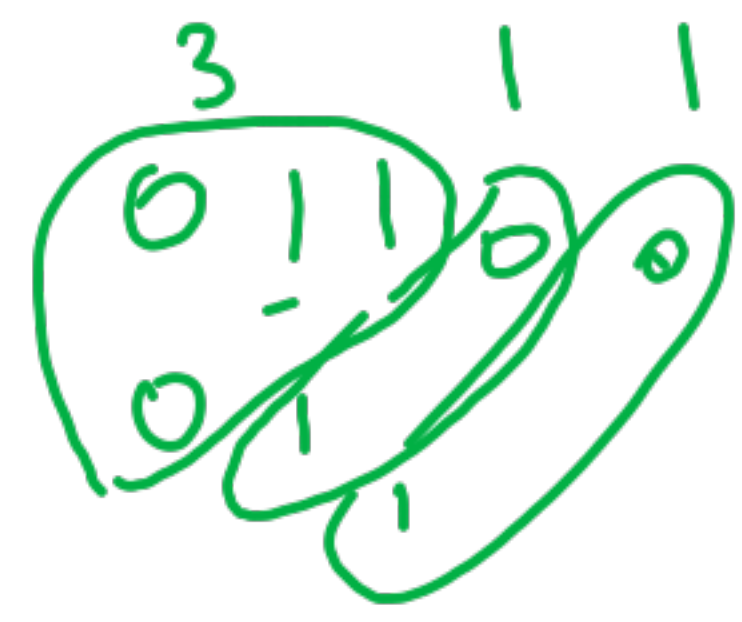
Solution:

$$i_1, i_2, \dots, i_m$$

↑
3
2
1
sequence

identical

x_{i_1}	x_{i_2}	x_{i_m}
y_{i_1}	y_{i_2}	y_{i_m}



Smallest Solution is length 75

Quiz 2
(Oct 15)
Wed

Assignment 4 on Moodle

90% started slides

Future

- 1 - TM $\equiv$ Type 0
 - 2 - Rec. Fn. theory
 - 3 - $f(n): N \rightarrow N$
 - 4 - λ -Calculus
 - 5 - Term Rewriting
- =
- NP-completeness

TMs

Variants

tracks

Assembly
lane $\leftrightarrow$

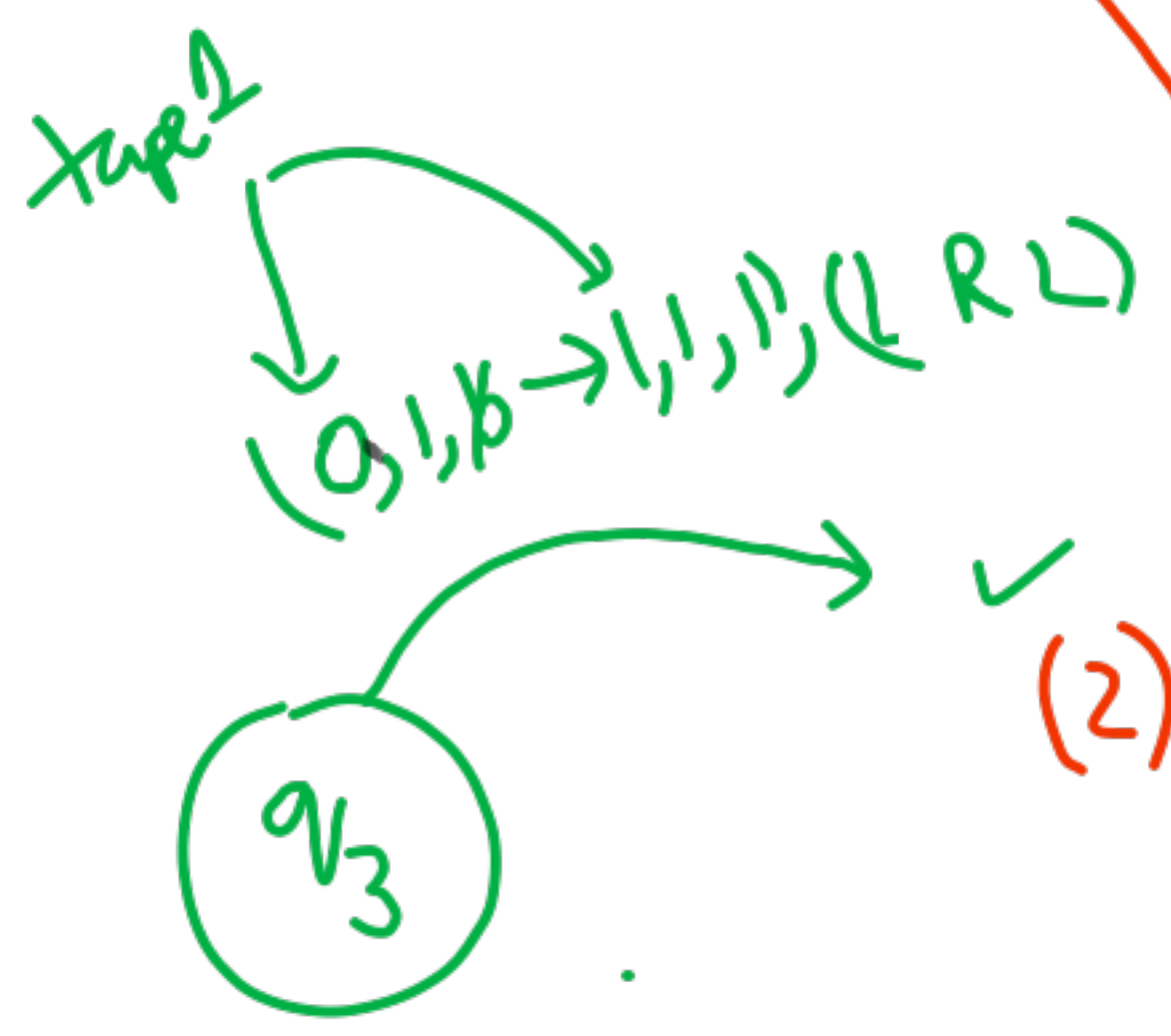
$\langle Q, \Sigma, \Gamma, \delta, q_0, \emptyset, F \rangle$

Variants

③ Non-deterministic



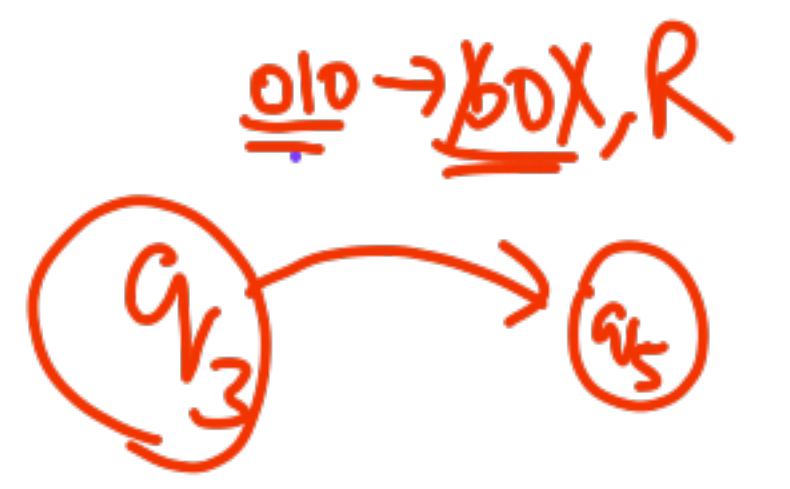
④ 1-way tape ✓



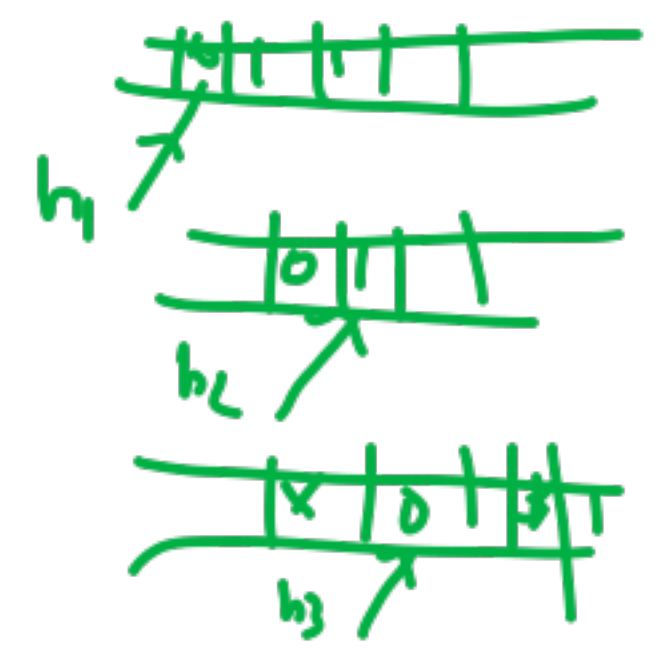
(1) Multi-track

	x	0	b
	y	1	1
	z	0	0

head



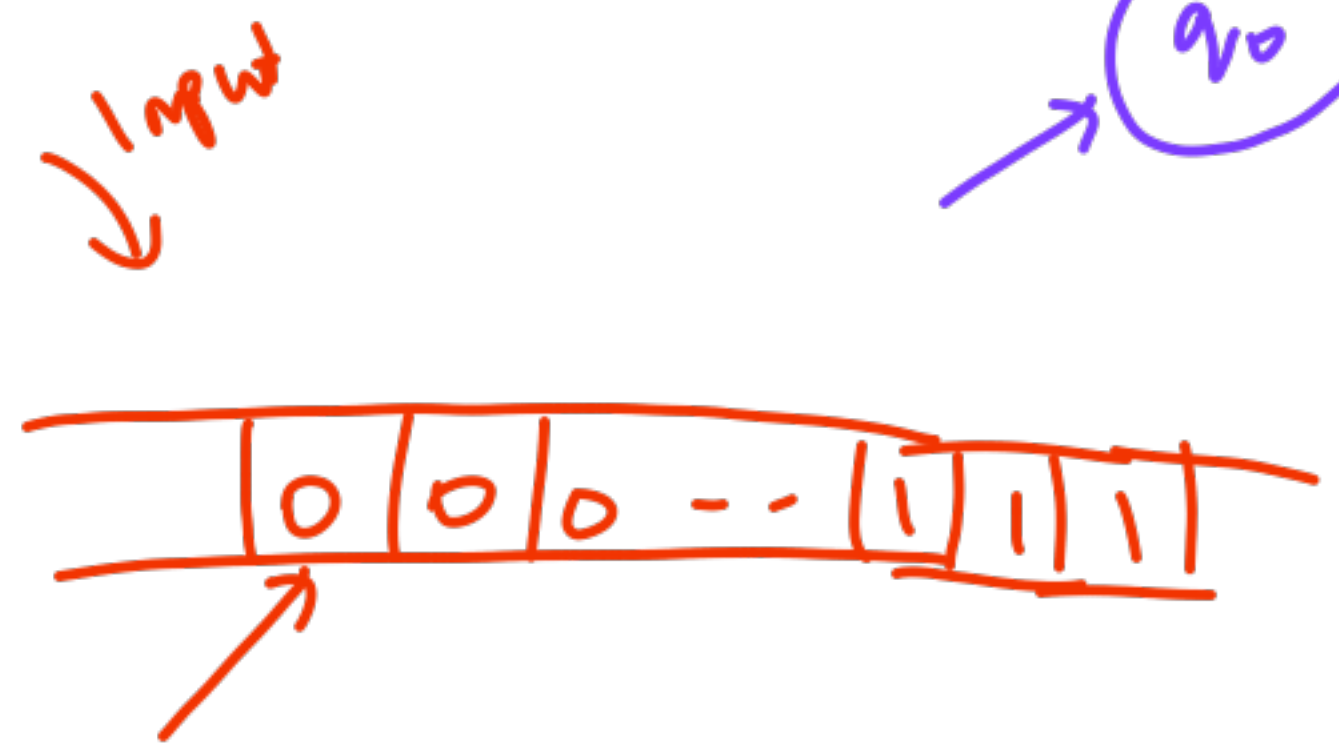
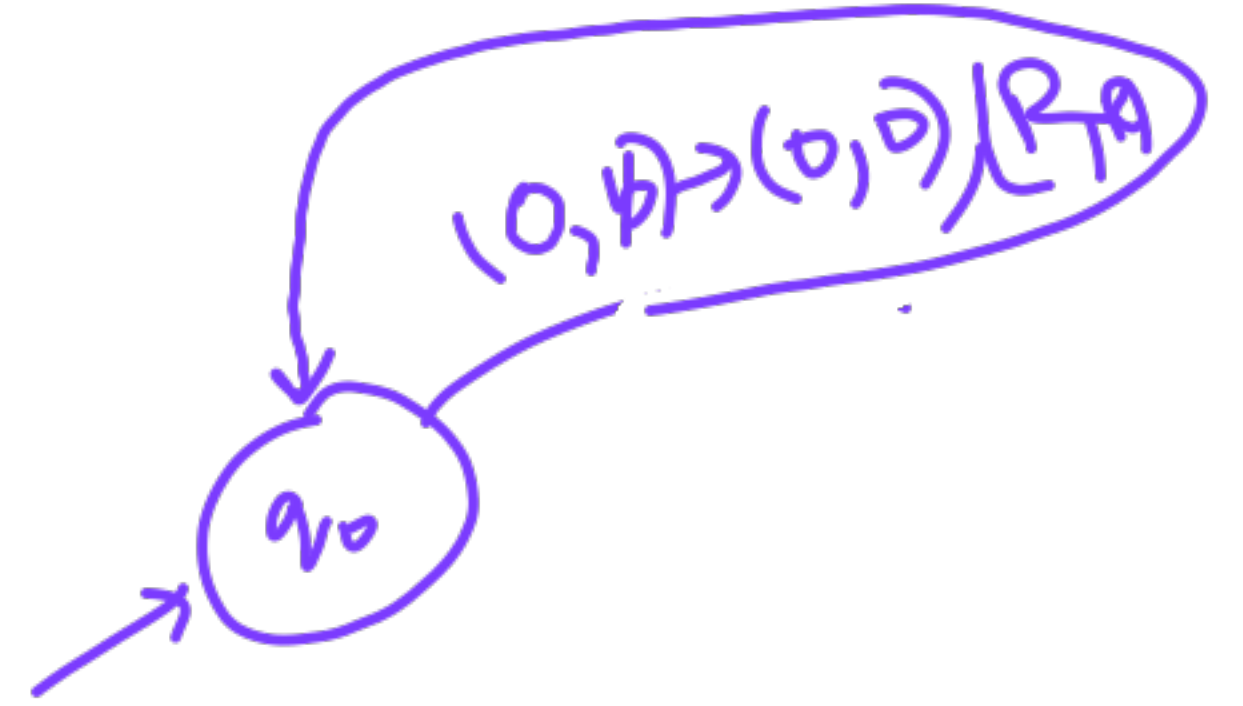
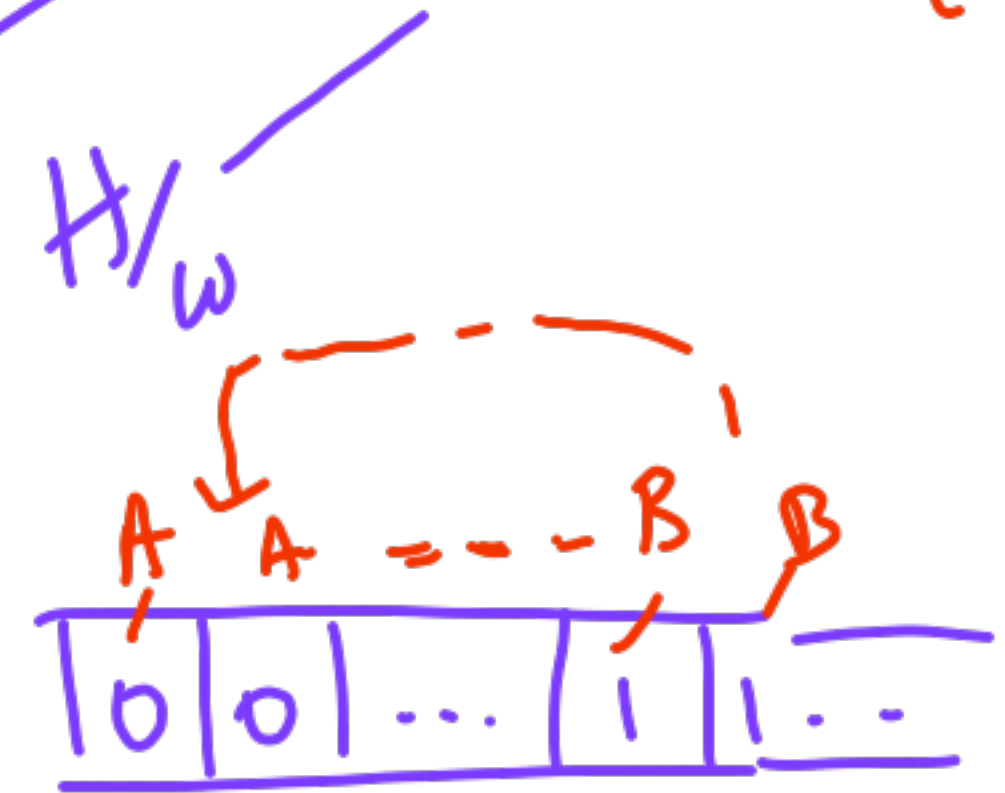
(2) Multi-tape



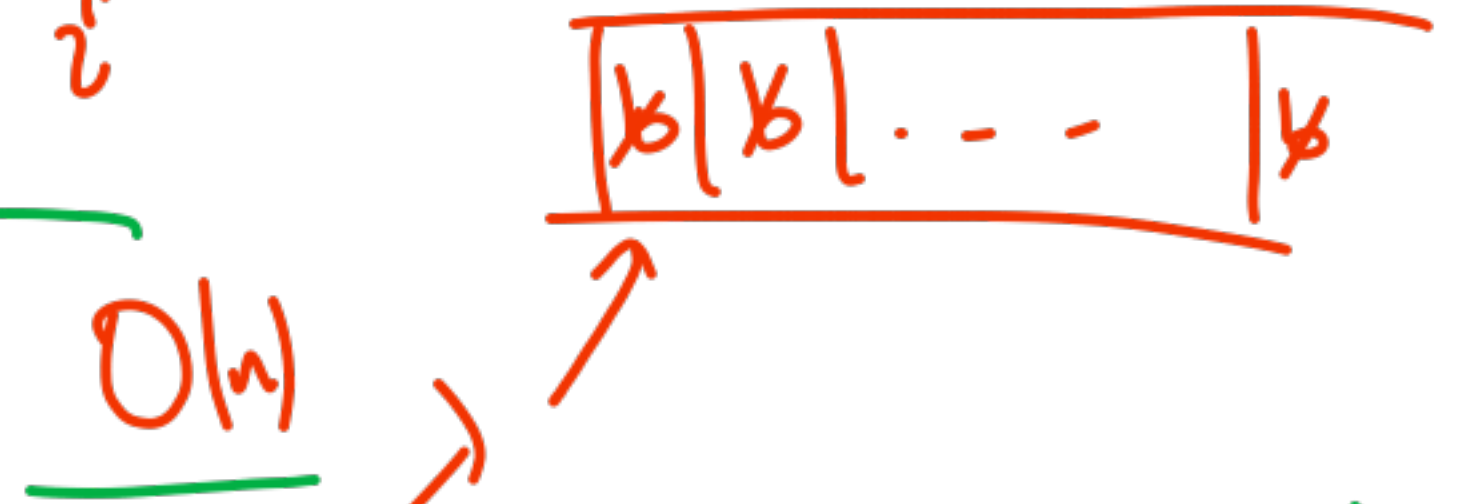
Turing machine. 10

$$L = \{ 0^n 1^n \mid n \geq 1 \}$$

✓ 2-tape

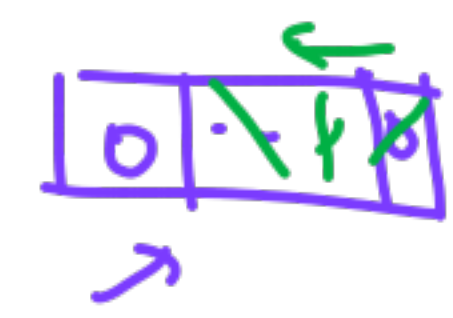
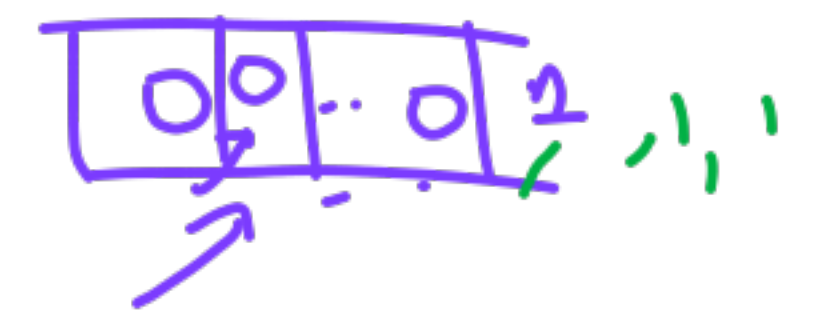


$O(n^2)$

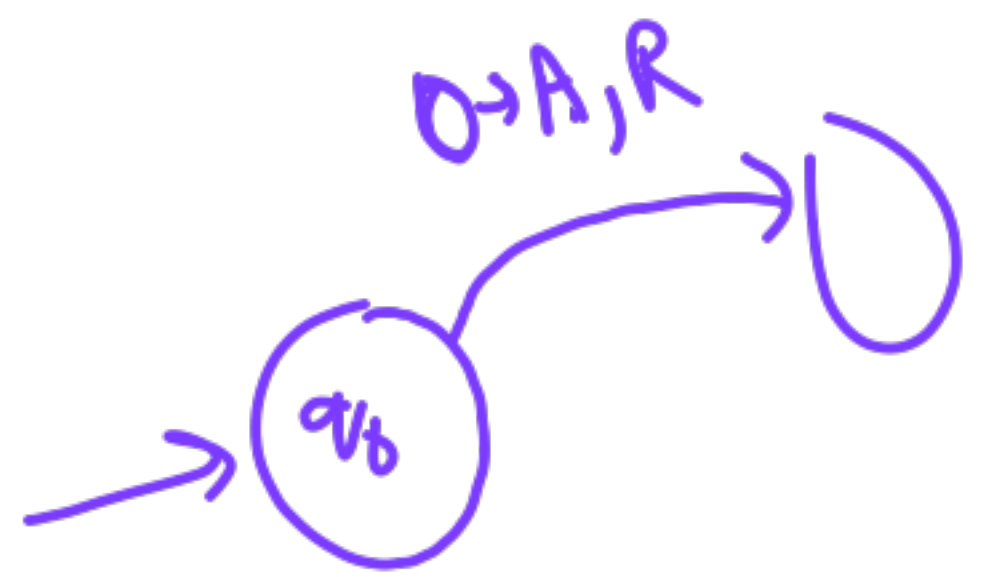
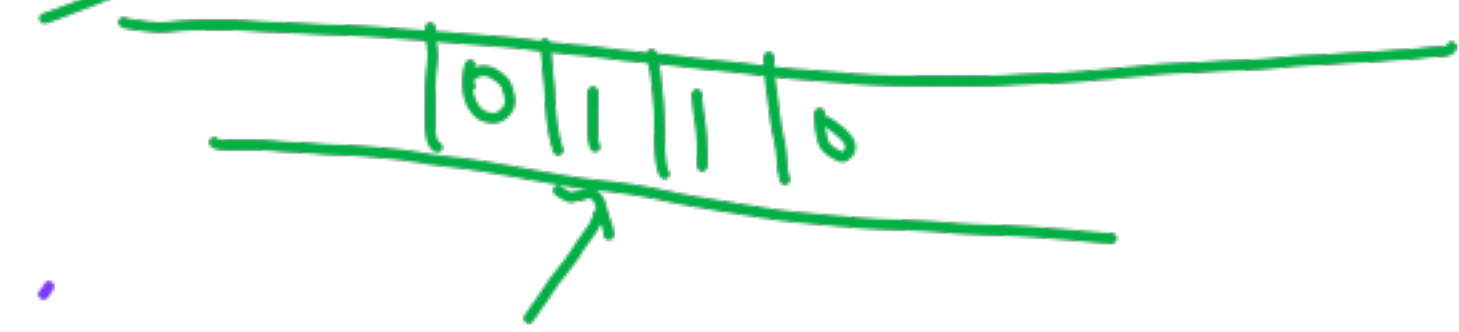


$O(n)$
work
easier

Idea



Shift & right



No extra-power
for any of
the variants

Simple-type JM
deterministic

prots
(very technical)

Can convert
systematically
to equivalent

can
mimic
all
variants

$a_j \rightarrow a_x / d_m$

j -th tape symbol $\rightarrow x$ -th symbol $\rightarrow m$ -th dir $\rightarrow p$



Model of
a (ception)

Encoding
machine begins

$0^i | 0^j | 0^l | 0^m | 0^k ||$

move 2
move 3
||
||

- ① Accept states
 $q_0 w + \dots + \alpha q_f \beta$ $q_f \in F$
- ② Halt $q_0 w + \dots + \alpha q_i \beta$ $\rightarrow$ no moves possible

H/w

DFA

Reg
Dyn. Prog

Languages

$L_d \neq \emptyset$

Rec. Enumerable $\rightarrow$

L_u

Recursive $\rightarrow$

L is rec. $\Leftrightarrow \exists$ TM M

that halts on all w

and is in accept state if $w \in L$
in reject state if $w \notin L$

L

Closure

Union

Inter

Cat

Rev

Difference

Complement

L is r.e. $\Leftrightarrow$ there is a TM M
Such that $L(M) = L$

M accepts all $w \in L$

